

Demostracion de Producto interior

1) $\forall v \neq 0_v \cdot \langle v, v \rangle > 0$

2) $\langle v, v \rangle = 0 \Leftrightarrow v = 0_v$

3) $\langle Kv, w \rangle = K \langle v, w \rangle, K \in \mathbb{R}$

4) $\langle v, w \rangle = \langle w, v \rangle$

ortogonalidad

$w^\perp = \{v \in V \mid \langle v, w \rangle = 0, \forall w \in W\}$ $\longrightarrow$

Norma

$\|u\| = \sqrt{\langle u, u \rangle} \longrightarrow$

Gram-Schmidt $\rightarrow$

Ej. Considera el conjunto $B = \{ \overset{v_1}{(1, 1, 1)}, \overset{v_2}{(1, 1, 4)}, \overset{v_3}{(1, 3, 1)} \}$ de $\mathbb{R}^3$.
Encuentra un conjunto ortogonal, ordenado al pararse la ortogonalidad de Gram-Schmidt.

B₁ conjunto ortogonal $\{u_1, u_2, u_3\}$

$u_1 = v_1 = (1, 1, 1)$
 $\|u_1\|^2 = 1^2 + 1^2 + 1^2 = 3$
 $\hat{u}_1 = \frac{u_1}{\|u_1\|} = \frac{1}{\sqrt{3}}(1, 1, 1)$

$u_2 = v_2 - \frac{\langle v_2, \hat{u}_1 \rangle}{\langle \hat{u}_1, \hat{u}_1 \rangle} \hat{u}_1 = (1, 1, 4) - \frac{\langle (1, 1, 4), \frac{1}{\sqrt{3}}(1, 1, 1) \rangle}{\langle \frac{1}{\sqrt{3}}(1, 1, 1), \frac{1}{\sqrt{3}}(1, 1, 1) \rangle} \cdot \frac{1}{\sqrt{3}}(1, 1, 1)$
 $= (1, 1, 4) - \frac{1+1+4}{3} \cdot \frac{1}{\sqrt{3}}(1, 1, 1) = (1, 1, 4) - \frac{6}{3} \cdot \frac{1}{\sqrt{3}}(1, 1, 1) = (1, 1, 4) - 2\sqrt{3} \cdot \frac{1}{\sqrt{3}}(1, 1, 1) = (1, 1, 4) - 2(1, 1, 1) = (-1, 0, 2)$
 $\|u_2\|^2 = (-1)^2 + 0^2 + 2^2 = 5$
 $\hat{u}_2 = \frac{u_2}{\|u_2\|} = \frac{1}{\sqrt{5}}(-1, 0, 2)$

$u_3 = v_3 - \frac{\langle v_3, \hat{u}_1 \rangle}{\langle \hat{u}_1, \hat{u}_1 \rangle} \hat{u}_1 - \frac{\langle v_3, \hat{u}_2 \rangle}{\langle \hat{u}_2, \hat{u}_2 \rangle} \hat{u}_2 = (1, 3, 1) - \frac{\langle (1, 3, 1), \frac{1}{\sqrt{3}}(1, 1, 1) \rangle}{\langle \frac{1}{\sqrt{3}}(1, 1, 1), \frac{1}{\sqrt{3}}(1, 1, 1) \rangle} \cdot \frac{1}{\sqrt{3}}(1, 1, 1) - \frac{\langle (1, 3, 1), \frac{1}{\sqrt{5}}(-1, 0, 2) \rangle}{\langle \frac{1}{\sqrt{5}}(-1, 0, 2), \frac{1}{\sqrt{5}}(-1, 0, 2) \rangle} \cdot \frac{1}{\sqrt{5}}(-1, 0, 2)$
 $= (1, 3, 1) - \frac{1+3+1}{3} \cdot \frac{1}{\sqrt{3}}(1, 1, 1) - \frac{(-1+0+2)}{5} \cdot \frac{1}{\sqrt{5}}(-1, 0, 2) = (1, 3, 1) - \frac{5}{3} \cdot \frac{1}{\sqrt{3}}(1, 1, 1) - \frac{1}{5} \cdot \frac{1}{\sqrt{5}}(-1, 0, 2)$
 $= (1, 3, 1) - \frac{5}{3\sqrt{3}}(1, 1, 1) - \frac{1}{5\sqrt{5}}(-1, 0, 2)$
 $= (1, 3, 1) - \frac{5}{3\sqrt{3}}(1, 1, 1) + \frac{1}{5\sqrt{5}}(1, 0, -2)$
 $= (1 - \frac{5}{3\sqrt{3}}, 3 - \frac{5}{3\sqrt{3}}, 1 - \frac{5}{3\sqrt{3}}) + \frac{1}{5\sqrt{5}}(1, 0, -2)$
 $= (\frac{3\sqrt{3}-5}{3\sqrt{3}}, \frac{9\sqrt{3}-5}{3\sqrt{3}}, \frac{3\sqrt{3}-5}{3\sqrt{3}}) + \frac{1}{5\sqrt{5}}(1, 0, -2)$
 $= \frac{1}{3\sqrt{3}}(3\sqrt{3}-5, 9\sqrt{3}-5, 3\sqrt{3}-5) + \frac{1}{5\sqrt{5}}(1, 0, -2)$
 $\hat{u}_3 = \frac{u_3}{\|u_3\|} = \frac{1}{\sqrt{10}}(\frac{3\sqrt{3}-5}{\sqrt{3}}, \frac{9\sqrt{3}-5}{\sqrt{3}}, \frac{3\sqrt{3}-5}{\sqrt{3}}) + \frac{1}{5\sqrt{5}}(1, 0, -2)$ es un conjunto ortogonal de $\mathbb{R}^3$

Ej. Sean $u = (1, 1, 1)$ y $v = (1, 1, 1)$, ϕ es el producto interno para ambos vectores en $\mathbb{R}^3$.

A) Construye ϕ^*
 B) determina una base $\mathcal{B}$ de $\mathbb{R}^3$

$\phi^* = \begin{pmatrix} \langle u, u \rangle & \langle u, v \rangle \\ \langle v, u \rangle & \langle v, v \rangle \end{pmatrix} = \begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix}$
 Como $\begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix} \neq 0$, entonces es L.T. (v)

De $u, v \in B = \{(1, 1, 1), (1, 1, 1)\}$ es base de $\mathbb{R}^3$
 $\phi^* = \begin{pmatrix} \langle (1, 1, 1), (1, 1, 1) \rangle & \langle (1, 1, 1), (1, 1, 1) \rangle \\ \langle (1, 1, 1), (1, 1, 1) \rangle & \langle (1, 1, 1), (1, 1, 1) \rangle \end{pmatrix} = \begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix}$

Si $u, v \in \mathbb{R}^3$, de forma:

$\langle u, v \rangle = \langle v, u \rangle$
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$u, v \in \mathbb{R}^3$

$\mathcal{M}(\mathcal{R}) = \text{tr}(B^c \cdot A)$

$\mathcal{R} = \langle (x, y), (v) \rangle$
 $\mathcal{G} = \langle (x, y, z), (1, 0, 1) \rangle$
 $\Rightarrow x + z = 0$

$\mathcal{P}_{\mathcal{R}^\perp} = \text{Id}$ hay

$\longrightarrow$ Base ortogonal: $\text{ortonormal} = \frac{B_0}{\|B_0\|}$

Transformacion Lineal

1) $\forall v_1, v_2 \in V: T(v_1 + v_2) = T(v_1) + T(v_2)$

2) $\forall K \in \mathbb{R}, \forall v \in V: T(Kv) = KT(v)$

2) Sean $T: \mathbb{R}^2 \rightarrow \mathcal{P}(\mathbb{R})$
 $(a, b) \rightarrow T(a, b) = a_1 + b$

$\hat{C} B_0 \in T \cup \mathcal{L}?$

1) $\forall v_1, v_2 \in \mathbb{R}^2: T(v_1 + v_2) = T(v_1) + T(v_2)$

$v_1 = (a_1, b_1) \in \mathbb{R}^2, v_2 = (a_2, b_2) \in \mathbb{R}^2$

$T(v_1 + v_2) = T((a_1, b_1) + (a_2, b_2))$
 $= T((a_1 + a_2, b_1 + b_2))$
 $= (a_1 + a_2) + (b_1 + b_2)$
 $= a_1 + a_2 + b_1 + b_2$
 $= (a_1 + b_1) + (a_2 + b_2)$
 $= T((a_1, b_1)) + T((a_2, b_2))$
 $= T(v_1) + T(v_2)$

2) $\forall K \in \mathbb{R}, \forall v \in \mathbb{R}^2: T(Kv) = KT(v)$

$v = (a, b) \in \mathbb{R}^2$

$T(Kv) = T(K(a, b))$
 $= T((Ka, Kb))$
 $= Ka + Kb$
 $= K(a + b)$
 $= K(T((a, b)))$
 $= KT(v)$

De $1, 2 \rightarrow T \in \mathcal{L}$

Núcleo e imagen de una T.L

$$\text{Ker}(\tau) = \{v \in V \mid \tau(v) = \theta_w\}$$

$$\text{Im}(\tau) = \{w \in W \mid \exists v \in V \wedge \tau(v) = w\}$$

Ej: Sea $\tau: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ salida llegada
 $(x, y) \mapsto \tau(x, y) = (x, x+y)$; una T.L. Caracterice $\text{Ker}(\tau)$ e $\text{Im}(\tau)$

$$\begin{aligned} \text{Ker}(\tau) &= \{v \in \mathbb{R}^2 \mid \tau(v) = \theta_{\mathbb{R}^2}\} \\ &= \{(x, y) \in \mathbb{R}^2 \mid \tau(x, y) = (0, 0)\} \\ &\quad (x, x+y) = (0, 0) \end{aligned}$$

$$\begin{aligned} x &= 0 \Rightarrow x+y=0 \\ x+y &= 0 \end{aligned}$$

$\text{Ker}(\tau) = \{(0, 0)\}$ → $\dim(\text{Ker}(\tau)) = 0$; ya que su único vector es el nulo
 $\therefore \tau$ es una T.L. inyectiva

$$\begin{aligned} \text{Im}(\tau) &= \{(a, b) \in \mathbb{R}^2 \mid \exists (x, y) \in \mathbb{R}^2, \tau(x, y) = (a, b) \\ &\quad (x, x+y) = (a, b) \end{aligned}$$

$$\begin{aligned} x &= a \\ x+y &= b \rightarrow y = b-a \end{aligned} \quad \left. \begin{array}{l} x=a \\ x+y=b \end{array} \right\} \text{Infinitas sol.}$$

$$\text{Im}(\tau) = \mathbb{R}^2 \quad \therefore \tau \text{ es sobreyectiva; ya que } \text{Im}(\tau) = \mathbb{R}^2 = \mathbb{R}^2$$

$$\dim(\text{Ker}(\tau)) = 0, \dim(\mathbb{R}^2) = 2$$

$$\begin{array}{cccc} m(\tau) + n(\tau) & = & \dim(V) \\ 0 & + & 2 & = & 2 \end{array}$$

Matriz asociada a una T.L

$$\text{Ej: } \tau(V_{B_1}) = \tau(x, y, z) = \alpha(V_{B_2}) + \beta(V_{B_2})$$

Ej: Sea $\tau: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ lineal y $B_1 = \{ \overset{v_1}{(1, 0, 0)}, \overset{v_2}{(1, 1, 0)}, \overset{v_3}{(1, 1, 1)} \}$, $B_2 = \{(0, 1), (1, 1)\}$
 Bases de $\mathbb{R}^3$ y $\mathbb{R}^2$ respectivamente. τ está definida por: $\tau(x, y, z) = (x+y, y+z)$
 Calcular $[\tau]_{B_2}^{B_1}$

$\tau(\overset{v_1}{(1, 0, 0)}) = (1, 0) = \alpha(0, 1) + \beta(1, 1)$; $\beta = 1, \alpha + \beta = 0 \rightarrow \alpha = -1$
 $\tau(\overset{v_2}{(1, 1, 0)}) = (2, 1) = \alpha_1(0, 1) + \beta_1(1, 1)$; $\beta_1 = 2, \alpha_1 + \beta_1 = 1 \rightarrow \alpha_1 = -1$
 $\tau(\overset{v_3}{(1, 1, 1)}) = (2, 2) = \alpha_2(0, 1) + \beta_2(1, 1)$; $\beta_2 = 2, \alpha_2 + \beta_2 = 2 \rightarrow \alpha_2 = 0$

$$[\tau]_{B_2}^{B_1} = \begin{bmatrix} -1 & -1 & 0 \\ 1 & 2 & 2 \end{bmatrix}_{2 \times 3}$$

vector de coordenadas

$$[T]_{B_1}^{B_2} [v]_{B_1} = [T(v)]_{B_2}$$

Ej: Sea $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ lineal, $B_1 = \{(1,0,0), (1,1,0), (1,1,1)\}$ y $B_2 = \{(0,1), (1,1)\}$ bases de $\mathbb{R}^3$ y $\mathbb{R}^2$ respectivamente. Sea además $[T]_{B_1}^{B_2} = \begin{bmatrix} -1 & -1 & 0 \\ 1 & 2 & 2 \end{bmatrix}$.
Calcule $T(x, y, z)$

$$[T]_{B_1}^{B_2} [v]_{B_1} = [T(v)]_{B_2}$$

vector de coordenadas

$$v = (x, y, z)$$

v_{B_1}

$$(x, y, z) = \alpha(1,0,0) + \beta(1,1,0) + \gamma(1,1,1)$$

$$\left. \begin{array}{l} \alpha + \beta + \gamma = x \\ \beta + \gamma = y \\ \gamma = z \end{array} \right\} \Rightarrow [v]_{B_1} = \begin{bmatrix} x-y \\ y-z \\ z \end{bmatrix}$$

$$[T]_{B_1}^{B_2} [v]_{B_1} = \begin{bmatrix} -1 & -1 & 0 \\ 1 & 2 & 2 \end{bmatrix} \begin{bmatrix} x-y \\ y-z \\ z \end{bmatrix} \Rightarrow \begin{bmatrix} -x + y - y + z \\ x - y + 2y - 2z + 2z \end{bmatrix} \Rightarrow \begin{bmatrix} -x + z \\ x + y \end{bmatrix} = [T(x, y, z)]_{B_2}$$

$T(x, y, z)$ v_{B_2} $T(v)$ v_{B_2}

$$T(x, y, z) = (-x + z)(0,1) + (x + y)(1,1) \rightarrow (0, -x + z) + (x + y, x + y) = (x + y, z + y)$$

$$T(x, y, z) = (x + y, y + z)$$

→ sacar $T(x, y, z)$

Calcular T^{-1}

b) calcule T^{-1}

$$Im(T) = \{(a, b) \in \mathbb{R}^2 / \exists (x, y) \in \mathbb{R}^2, T(x, y) = (a, b)\}$$

$$\begin{array}{l} T(x, y) = (a, b) \\ (x, y) = T^{-1}(a, b) \end{array} \quad / \quad T^{-1}$$

$$\begin{array}{l} x - 2y = a \\ x + y = b \end{array} \rightarrow \begin{array}{l} x - 2y = a \\ x + y = b \end{array} \xrightarrow{y = \frac{b-a}{3}} \rightarrow x = \frac{2}{3}b + \frac{a}{3}$$

$$\therefore T^{-1}(a, b) = \left(\frac{2b+a}{3}, \frac{b-a}{3} \right)$$

Cuando es el caso de una matriz asociada hay que usar la identidad y operaciones elementales